

The Salem boundary for substitution spectral measures: scoped later-literature answers to arXiv:1305.7373

Literature status note — agent_lane_12

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Status. *Literature-already-answered partial subcases; the exact log-Hölder bound remains open.*

Original question

Bufetov and Solomyak [1] consider a primitive aperiodic substitution with irreducible substitution matrix and Salem Perron–Frobenius eigenvalue α . On page 15 in Section 4, immediately after Theorem 4.1, they say that they do not know whether almost-every suspension flow has Hölder spectral estimates in this borderline case and expect failure. In Section 5, immediately after Theorem 5.1 and equation (5.1), they ask whether the self-similar flow nevertheless satisfies (page 26, Remark 5.3)

$$\sigma_a([\omega - r, \omega + r]) \leq C_B(\log(1/r))^{-\gamma}, \quad |\omega| \in [B^{-1}, B], \quad r \downarrow 0. \quad (\text{Q})$$

Explicit later answers

Marshall-Maldonado [2] states that the main objective is the Salem question raised in [1]. Theorems 1.1–1.2 prove Hölder local bounds for the self-similar flow at algebraic spectral parameters $\omega \in \mathbb{Q}(\alpha)$. The exponent is controlled by arithmetic parameters, but the admissible starting radius depends on ω ; hence this is not the uniform estimate (Q).

Marshall-Maldonado and Solomyak [3] give the following current boundary. Their results concern Lip-cylindrical mean-zero functions; the letter-cylinder measures in [1] are included after subtracting the constant component away from frequency zero.

Result in [3]	Scope	Conclusion
Theorem 2.3 (pp. 6–7)	Lebesgue-a.e. roof vector	Uniform spectral modulus $\exp[-\beta L_3/L_4]$, where L_3 and L_4 are the relevant third and fourth iterated logarithms; quantitative weak mixing follows.
Theorem 2.4 (p. 7)	Every roof vector; almost every Lip-cylindrical test function	Every uniform Hölder bound Cr^η , $\eta > 0$, fails along a sequence of scales. This confirms the expected failure of the Section 4 Hölder regime.
Theorem A.2 (p. 18)	The self-similar roof	Uniformly in $\omega \in \mathbb{R}$, $\sigma_f(B_r(\omega)) \leq a_\alpha \exp[-c_\alpha \log^*(r^{-1})]$, with the same quantitative weak-mixing rate.

What remains open

Theorem A.2 is the first uniform quantitative bound for the self-similar Salem case, but $\exp[-c \log^*(1/r)]$ is much weaker than every negative power of $\log(1/r)$. The authors of [3] explicitly say that a log-Hölder rate seems plausible and is not known. Thus the later literature answers substantial parts of the extracted Salem question but neither proves nor disproves (Q).

The arithmetic obstruction is visible in [3, Proposition A.3]: the escape scales satisfy $k_{i+1} = L^{k_i}$, giving only $\sum_{n < N} \|w\alpha^n\|_{\mathbb{R}/\mathbb{Z}}^2 \gtrsim \log^* N$. A proof of (Q) would require a qualitatively denser uniform escape estimate.

Provenance and search evidence

Both supporting papers explicitly cite [1] as the source of the Salem question, so this is a direct literature answer rather than an agent-inferred reformulation. Searches covered the exact source sentence, “Salem substitution suspension flow spectral measure”, the three arXiv identifiers, and the cited 2025 discrete-action sequel. No later proof or disproof of (Q) was found.

References

- [1] A. I. Bufetov and B. Solomyak, *On the modulus of continuity for spectral measures in substitution dynamics*, Adv. Math. 260 (2014), 84–129; arXiv:1305.7373.
- [2] J. Marshall-Maldonado, *Modulus of continuity for spectral measures of suspension flows over Salem type substitutions*, Israel J. Math. 263 (2024), 657–685; arXiv:2009.13607.
- [3] J. Marshall-Maldonado and B. Solomyak, *Quantitative weak mixing for typical Salem substitution suspension flows*, arXiv:2601.15035 (2026), Theorems 2.3, 2.4, and A.2.